Scenario 1: Identifying Trend
You are given a dataset showing daily temperature for the past year. You notice the values generally increase over time. How do you confirm the presence of a trend?

Stepwise Solution:

1. Plot the data → Use a line plot to observe general upward/downward movement.
2. Check rolling mean → Use `.rolling(window=30).mean()` to smooth the series.
3. Use statistical test → Apply Augmented Dickey-Fuller (ADF) test:
   If p-value > 0.05 → non-stationary → likely trend exists.
   Conclusion: Data has a trend, confirmed by plot and ADF test.

Scenario 2: Detecting Seasonality
You are analyzing monthly electricity usage and notice similar spikes every summer. How do you confirm seasonality?

Stepwise Solution:

1. Plot the time series → Visually check for repeating patterns.
2. Use seasonal decomposition
3. Check autocorrelation plot (ACF)

   Conclusion: Data has yearly seasonality.

Scenario 3: Making Data Stationary

Question:
Your time series model gives poor results. The data shows trend and seasonality. How do you prepare it for ARIMA?

Stepwise Solution:

1. Check stationarity → Use ADF test.
2. Remove trend → Apply first-order differencing:
   df_diff = df.diff().dropna(),
3. Remove seasonality → Apply seasonal differencing:
   df_seasonal_diff = df.diff(12).dropna()
4. Re-check ADF test on transformed data.

   Conclusion: Differencing removes non-stationarity, making it suitable for ARIMA.

Scenario 4: Choosing Model for No Seasonality
Your time series data (e.g., daily website visits) has no visible seasonality but has a clear trend. Which model would you choose?

Stepwise Solution:

1. Plot the data → Confirm no repeating seasonal pattern.

2. Use ADF test → To check stationery.

3. If trend present and no seasonality:

4. Use ARIMA model (not SARIMA or Holt-Winters).

5. Use auto_arima or ACF/PACF to find parameters.

Conclusion: Choose ARIMA for trend-only time series.

Scenario 5: Evaluating Forecast Performance
You forecast sales for 3 months and want to check how accurate it is. What steps do you follow?

Stepwise Solution:

1. Compare forecast vs actual values.
2. Calculate error metrics:
   MAE (Mean Absolute Error)
   RMSE (Root Mean Squared Error)
   MAPE (Mean Absolute Percentage Error)
3. Interpret results:
   Lower values = better accuracy.
   Conclusion: Use error metrics to evaluate forecast quality.

Scenario 6: Sudden Spike in Data

You're analyzing daily water consumption. One day, the value suddenly jumps very high. What will you do?

Stepwise Solution:

1. Check if it's a real event – Was there a water leak or a festival?
2. If real → Keep the value (it's part of the pattern).

If error or outlier → You can:
Replace it with the average of nearby values.
Or use smoothing methods like rolling average.

Scenario 7: Sales Goes Up Every December

You're analyzing a store's monthly sales. Sales always increase in December. What does this indicate?

Step-by-step Solution:

1. Check if this happens every year → Yes.
2. This means there is seasonality → A pattern that repeats every year.
3. Model to use: Use models that handle seasonality like:
   Holt-Winters
   SARIMA
   Facebook Prophet
   Conclusion: December sales pattern = seasonality.

Scenario 8: Forecasting with Missing Values

You are forecasting stock prices, but a few days of data are missing. What should you do?

Step-by-step Solution:

1. Check how much data is missing:
   If only a few values, you can fill with average of nearby days.
2. Don't drop rows unless the missing part is too large.
3. Use forward fill or interpolation:
   df.fillna(method='ffill', inplace=True)

   Conclusion: Handle missing values before building the model.